Tianyu Wang (Ph.D.)

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Education

2020.09 – 2025.06 **Zhi Yuan Honorary Ph.D., Shanghai Jiao Tong University**
University of Michigan-Shanghai Jiao Tong University Joint Institute
Control Science and Engineering
GPA: 3.75 / 4.0

2023.02 – 2024.02 **Visiting Scholar, KTH Royal Institute of Technology**
School of Industrial Engineering and Management
Production Engineering
Supervisor: Lihui Wang (20 most influential professors in smart manufacturing)

2019.01 – 2019.02 **Winter Program, Massachusetts Institute of Technology**
Department of Electrical Engineering and Computer Science
First Rank of Overall Grade

2016.09 – 2020.08 **Bachelor, Shanghai Jiao Tong University**
University of Michigan-Shanghai Jiao Tong University Joint Institute
Electrical and Computer Engineering
GPA: 3.65 / 4.0

Work Experience

2025.11 – NOW **Postdoctoral Researcher on AI-Empowered Manufacturing and Human-Robot Collaboration**
KTH Royal Institute of Technology, Stockholm, Sweden
Department of Production Engineering, Production Systems & Automation Division
Supervisors: Xi Vincent Wang and Lihui Wang

- Work on AI for manufacturing, with a focus on large language model agents and agentic AI for autonomous design, reconfiguration, and governance of next-generation manufacturing systems.
- Work on adaptive human-robot collaboration, including world models for long-horizon human anticipation, operator-aware personalization, and safety-aware symbolic robot policies for assembly tasks.
- Work on physical AI for industrial robots, integrating foundation models, dynamical-systems-based decoders, and digital twins to enable robust closed-loop perception, planning, and control.

Work Experience (continued)

2023.02 – 2024.02 **Researcher on Human-Robot Collaboration towards Industry 5.0**
KTH Royal Institute of Technology, Stockholm, Sweden

- Work on vision-based human mesh recovery, human action recognition and human intention prediction with foundation models and multimodal data fusion.
- Work on high-fidelity digital twin of shared workspace including multiple humans and industrial robots, as well as human-aware robot learning and control.
- Collaborate with industry partners on EU project of HRC assembly with pilot cases.

2020.09 – 2025.10 **Researcher on Intelligent Manufacturing in Industry 4.0**
Shanghai Jiao Tong University, Shanghai, China

- Work on novel deep learning and continual learning-based methods for adaptive system identification and highly generalizable product quality prediction under concept drift.
- Work on human-inspired methods and large language models for effective knowledge consolidation among different operating conditions and production lines.

2018.04 – 2021.06 **Researcher on Vision-Based Action Quality Assessment**
Shanghai Jiao Tong University, Shanghai, China

- Work on deep learning algorithms for RGB video and kinematics-based real-time assessment of complex human body and hand actions.
- Collaborate with top hospitals on surgical skill assessment systems for VR training.

2018.10 – 2020.10 **Researcher on Smartphone User Data Mining**
Shanghai Jiao Tong University, Shanghai, China

- Work on machine learning algorithms to mine the multi-level user behavior from massive smartphone data featuring high dimension and high dispersion.
- Develop a big data platform with dedicated utilities for commercial data mining.

Selected Research

2025.11 – NOW **Robustness-by-Design World Model for Human-Robot Co-Assembly**
Human-Robot Collaboratory

- Develop a hybrid latent-structured world model predicting long-horizon human futures from multi-view RGB and skeletons, coupled to a symbolic robot policy that scores candidate actions over predicted rollouts.
- Achieve over 96% perception accuracy and four-fold policy invariance on occlusion-heavy scenarios, and add an operator-aware personalization layer that cuts per-trial task-time error by over 40%.

Selected Research (continued)

2025.11 – 2026.04

Contract-Based Production Autonomy for LLM-Driven Intent-Based Manufacturing
Production Systems & Automation Division

- Propose a five-layer governance framework aligned with Industry 5.0 human-centric principles that treats production autonomy as a contract-compliance problem, with LLMs serving as bounded semantic agents and a verification firewall enforcing safety and well-being constraints.
- Validate on two case studies using an Isaac Sim digital twin with BaSyx and OPC-UA interfaces, together with a no-LLM ablation, quantifying the complementary roles of LLMs and calibrated baselines under disturbances.

2025.10 – 2026.04

LLM Agents-Driven Layout Generation for Reconfigurable Manufacturing Systems
Production Systems & Automation Division

- Develop a multi-agent LLM workflow that translates high-level natural-language requirements of manufacturing systems into geometry-feasible reconfigurable layouts using a knowledge base of 3D machine models.
- Integrate Unity3D-based visualization and downstream optimization to enable rapid, human-in-the-loop layout iteration, with results published in CIRP Annals (2026).

2026.01 – NOW

Ghost Attractor Networks: Basin-Structured Dynamical Decoders for Closed-Loop Sequential Generation
Human-Robot Collaboratory

- Propose a theoretically derived dynamical decoder whose latent state evolves under a learned potential with drift, producing a basin-attractor structure by construction for multi-modal, constant-memory closed-loop sequential generation.
- Realize behavioral mode transitions as saddle-node bifurcations with ghost-attractor escape, enabling single-pass mode switching at deployment frequencies of 50–100 Hz without token accumulation.
- Develop a hierarchical phase-space decomposition separating first-order basin convergence from second-order proprioceptive refinement; manuscript under review at IEEE Transactions on Neural Networks and Learning Systems, with preprint available as arXiv:2606.18315.

2025.06 – 2026.03

GenAI-Augmented Industrial Robots for Sustainable Manufacturing
Human-Robot Collaboratory

- Propose a universal, modular framework integrating ROS2, large language models, and vision foundation models for programming-free robot control.
- Enable natural-language command understanding, semantic task planning, and motion execution from raw environmental images.
- Validate on two ABB industrial robots in an automation cell with unpredictable settings, advancing the sustainable use of legacy industrial robots for manufacturing; results published in Engineering (2026).

Selected Research (continued)

2024.10 – 2025.06 **Cross-Domain Multimodal Language Model for Industrial Soft Sensors**
Intelligent Manufacturing Research Lab & Human-Robot Collaboratory

- Develop a specialized encoder–decoder LM with self-attention-based fusion of numerical, categorical, and textual variables for predictive soft sensing.
- Integrate domain-specific pre-training, low-rank fine-tuning, and gradient-free test-time adaptation to handle non-stationarity such as sensor degradation.

2024.02 – 2024.09 **Brain-Inspired Product Quality Prediction under Variations**
Intelligent Manufacturing Research Lab

- Propose a dual-branch modeling method inspired by the neuroscience theory of complementary learning systems for the first time in the field of product quality prediction for complex manufacturing processes.
- Achieve the SOTA performance on three real-world manufacturing processes across different industries (food production, injection molding and coal-fired electricity generation) with sufficient data records (over 15,000 in total).

2024.04 – 2024.10 **Efficient Industrial Adaptation of Segment Anything Model**
Intelligent Manufacturing Research Lab

- Additional output tokens are inserted into the decoder part and lightweight adapters are attached to the encoder part, along with a information transfer bridge.
- Achieve superior performances on multiple industrial defect segmentation datasets with only 10+ training images such as CT and X-ray.

2023.06 – 2024.09 **Human-Centric Digital Twin for Industry 5.0 Work Cells**
Human-Robot Collaboratory

- Propose a systematic framework for rapid and precise construction of human-centric DT systems in human–robot collaborative environments.
- Generate customized mesh-based human models from short motion videos within one minute, incorporating rendered appearances for physical traits.
- Develop transformer-based networks for robust motion tracking and procedure recognition under occlusions and incomplete observations.

2023.02 – 2024.02 **Cross-Domain Few-Shot Multimodal Human Action Recognition**
Human-Robot Collaboratory

- Develop a few-shot learning model based on transformer and lightweight domain adapters under HRCA for data-efficient recognition and cross-domain fine-tuning.
- Design a hierarchical multimodal data fusion mechanism that leverages the context-aware motion information to mitigate the confusion of similar actions and various occlusions.

Selected Research (continued)

2023.01 – 2024.01 **Self-Adaptive Modeling of Multimode Continuous Processes**
Intelligent Manufacturing Research Lab

- Design a deep learning-based self-adaptation mechanism to improve the stability of the predictive model towards various changes in a long period of manufacturing.
- Develop a task-free knowledge consolidation strategy for the continual learning-based regularization to regulate the online update of predictive model.
- Investigate the largely unexplored aspect of proactive input data adaptation to deal with the non-stationary time series in an online manner.

2022.06 – 2023.03 **Continual Knowledge Transfer of Multiple Manufacturing Processes**
Intelligent Manufacturing Research Lab

- Explore the practical scenario of multiple distinct yet related manufacturing processes in a scaling modern factory for the digital twin-based decision making.
- Develop a continual knowledge transfer paradigm that addresses the existing issues of non-stationary process behavior regarding transfer learning and frequently changing tasks without clear boundaries regarding continual learning.

Research Publications

Journal Articles

1. **T. Wang**, S. Lu, T. Li, C. Cao, M. Li, Y. Huang, X. V. Wang, and L. Wang, "A bayesian refinement method incorporating geometric prior and LLM-guided hyperprior for measurement-grade overhang analysis in x-ray electrode inspection", *Advanced Engineering Informatics (AEI)*, vol. 77, p. 105 185, 2027. DOI: 10.1016/j.aei.2026.105185.
2. J. Yuan, **T. Wang**, X. V. Wang, L. Wang, J. Gao, M. Chiachío, C. Sbarufatti, and T. Li, "Prior information-aided severity-graded fatigue delamination prognostics in composites", *Advanced Engineering Informatics (AEI)*, vol. 77, p. 105 224, 2027. DOI: 10.1016/j.aei.2026.105224.
3. C. Cao, **T. Wang**, S. Lu, M. Li, Y. Huang, X. V. Wang, and L. Wang, "A segment anything model adaptation framework for battery visual inspection under complex radiographic imaging conditions", *Pattern Recognition*, vol. 180, p. 114 087, 2026. DOI: 10.1016/j.patcog.2026.114087.
4. Z. Chen, L. Peng, **T. Wang**, C. Zeng, Y. Pan, and J.-B. Wang, "Occlusion-aware visual object tracking with explicit temporal state modeling and dual-memory mechanism", *Pattern Recognition*, vol. 180, p. 114 155, 2026. DOI: 10.1016/j.patcog.2026.114155.
5. S. Grahn, **T. Wang**, X. V. Wang, and L. Wang, "From human–machine interaction to integration: A review of paradigms, principles, and prospects for a human-centric future", *Human-Machine Interaction*, vol. 1, no. 1, 2026, Corresponding Author. URL: https://www.sciltp.com/journals/hmi/articles/2608004893.
6. Z. Liu, **T. Wang**, R. Zhong, X. V. Wang, M. Li, and L. Wang, "Brain-inspired generative AI revives legacy industrial robots toward sustainable automation", *Engineering*, 2026. DOI: 10.1016/j.eng.2026.05.020.
7. J. Qu, J. Shen, W. Li, **T. Wang**, Y. Wang, R. Zheng, M. Li, and Z. Wang, "Diagnosing inconsistencies in battery energy storage systems: A framework integrating electrical, thermal, and aging perspectives", *Applied Energy*, vol. 405, p. 127 203, 2026. DOI: 10.1016/j.apenergy.2025.127203.

8 **T. Wang**, Z. Liu, X. V. Wang, and L. Wang, "LLM agents-driven layout generation for reconfigurable manufacturing systems", *CIRP Annals - Manufacturing Technology*, vol. 75, no. 1, pp. 577–581, 2026. DOI: 10.1016/j.cirp.2026.03.015.

9 **T. Wang**, M. Zhang, M. Lu, and M. Li, "Multi-timescale representation with adaptive routing for deep tabular learning under temporal shift", *Neural Networks*, vol. 199, p. 108 670, 2026. DOI: 10.1016/j.neunet.2026.108670.

10 **T. Wang**, M. Zhang, J. Qu, Z. Liu, M. Li, and X. V. Wang, "Multimodal knowledge-enhanced language model with online test-time adaptation for cross-domain industrial tabular prediction", *Engineering Applications of Artificial Intelligence (EAAI)*, vol. 178, p. 115 061, 2026. DOI: 10.1016/j.engappai.2026.115061.

11 T. Li, R. Zhong, **T. Wang**, J. Kronqvist, M. Romero, M. Xiao, and X. V. Wang, "Designing likelihood function under nuisance components in block particle filter", *Mechanical Systems and Signal Processing*, vol. 241, p. 113 595, 2025. DOI: 10.1016/j.ymssp.2025.113595.

12 Z. Liu, **T. Wang**, Z. Ji, W. Xu, L. Wang, and X. V. Wang, "Sequential human assembly and disassembly motions in human-robot coexisting environments", *Scientific Data*, vol. 12, no. 1, p. 1769, 2025.

13 F. Qi, **T. Wang**, R. Zheng, and M. Li, "A memetic and reflective evolution framework for automatic heuristic design using large language models", *Applied Sciences*, vol. 15, no. 15, p. 8735, 2025. DOI: 10.3390/app15158735.

14 J. Qu, **T. Wang**, Y. Wang, X. Li, M. Li, and R. Zheng, "An improved co-training architecture for lithium-ion batteries state of health estimation with semi-supervised learning", *Journal of Power Sources (JPS)*, vol. 643, p. 236 928, 2025.

15 **T. Wang**, Z. Hu, Y. Wang, M. Li, Z. Liu, and X. V. Wang, "A human-inspired slow-fast dual-branch method for product quality prediction of complex manufacturing processes with hierarchical variations", *Advanced Engineering Informatics (AEI)*, vol. 64, p. 102 967, 2025.

16 **T. Wang**, Z. Liu, L. Wang, M. Li, and X. V. Wang, "A design framework for high-fidelity human-centric digital twin of collaborative work cell in industry 5.0", *Journal of Manufacturing Systems (JMS)*, vol. 80, pp. 140–156, 2025.

17 **T. Wang**, R. Zheng, M. Li, C. Cai, S. Zhu, and Y. Lou, "Deep learning based self-adaptive modeling of multimode continuous manufacturing processes and its application to rotary drying process", *Journal of Intelligent Manufacturing (JIMS)*, vol. 36, no. 6, pp. 3887–3922, 2025.

18 C. Zhuang, J. Qu, **T. Wang**, L. Lin, Y. Bi, and M. Li, "Comprehensive MILP formulation and solution for simultaneous scheduling of machines and AGVs in a partitioned flexible manufacturing system", *Machines*, vol. 13, no. 6, p. 519, 2025. DOI: 10.3390/machines13060519.

19 **T. Wang**, Z. Liu, L. Wang, M. Li, and X. V. Wang, "Data-efficient multimodal human action recognition for proactive human–robot collaborative assembly: A cross-domain few-shot learning approach", *Robotics and Computer-Integrated Manufacturing (RCIM)*, vol. 89, p. 102 785, 2024.

20 Y. Gao, **T. Wang**, X. Zhou, M. Li, C. Zhang, P. Qin, Y. Yang, and L. Zhang, "An adaptive optimization method toward batch-wise variable set point of outlet moisture content for the tobacco drying process", *Drying Technology*, vol. 41, no. 13, pp. 2156–2170, 2023.

21 **T. Wang**, M. Li, R. Zheng, C. Cai, Y. Lou, and S. Zhu, "Towards continual knowledge transfer in modeling manufacturing processes under non-stationary data streams", *Applied Intelligence*, vol. 53, no. 23, pp. 29 393–29 413, 2023.

22 **T. Wang**, C. Cai, Y. Gao, M. Li, Y. Lou, and B. Kong, "Adaptive online learning for system identification and in-advance optimization under long feedback delay and concept drift in cigarette production", *Drying Technology*, vol. 40, no. 15, pp. 3340–3356, 2022.

23. M. Ding, **T. Wang**, and X. Wang, "Establishing smartphone user behavior model based on energy consumption data", *ACM Transactions on Knowledge Discovery from Data (TKDD)*, vol. 16, no. 2, pp. 1–40, 2021.

24. **T. Wang**, M. Jin, and M. Li, "Towards accurate and interpretable surgical skill assessment: A video-based method for skill score prediction and guiding feedback generation", *International Journal of Computer Assisted Radiology and Surgery (IJCARS)*, vol. 16, no. 9, pp. 1595–1605, 2021.

25. M. Wang, Y. Wang, **T. Wang**, L. Hou, and M. Li, "New approach for information security evaluation and management of IT systems in educational institutions", *Journal of Shanghai Jiaotong University (Science)*, vol. 25, no. 6, pp. 689–699, 2020. DOI: 10.1007/s12204-020-2231-y.

Conference Proceedings

1. C. Cao, **T. Wang**, M. Li, Y. Huang, J. Jiang, and S. Zhang, "An efficient segment anything model adaptation method for electrode overhang analysis in lithium-ion battery manufacturing", in *ASME Manufacturing Science and Engineering Conference (MSEC)*, Corresponding Author, vol. 89022, 2025, V002T13A003.

2. Z. Liu, S. Liu, **T. Wang**, L. Wang, and X. V. Wang, "Establishment and synchronisation of digital twins for multi-robot systems in manufacturing", in *Procedia CIRP*, vol. 134, 2025, pp. 419–424. DOI: 10.1016/j.procir.2025.02.152.

3. S. Lu, **T. Wang**, C. Cao, M. Li, Y. Huang, and S. Zhang, "A two-stage electrode refinement method with denoising diffusion for automated segmentation and overhang analysis on battery x-ray images", in *51st Annual Conference of the IEEE Industrial Electronics Society (IECON)*, Corresponding Author, 2025, pp. 1–8. DOI: 10.1109/IECON58223.2025.11221602.

4. J. Qu, M. Li, and **T. Wang**, "An order-free end-to-end multi-agv scheduling method based on safe deep q-network and asynchronous coordination", in *ASME International Design Engineering Technical Conferences and Computers and Information in Engineering Conference (IDETC-CIE)*, Corresponding Author, vol. 88353, 2024, V02BT02A023.

5. M. Lu, **T. Wang**, H. Zhu, and M. Li, "Hacl-net: Hierarchical attention and contrastive learning network for mri-based placenta accreta spectrum diagnosis", in *International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI)*, Corresponding Author, 2023, pp. 304–314.

6. **T. Wang**, M. Jin, J. Wang, Y. Wang, and M. Li, "Towards a data-driven method for rgb video-based hand action quality assessment in real time", in *Proceedings of the 35th Annual ACM Symposium on Applied Computing (SAC)*, 2020, pp. 2117–2120.

7. **T. Wang**, Y. Wang, and M. Li, "Towards accurate and interpretable surgical skill assessment: A video-based method incorporating recognized surgical gestures and skill levels", in *MICCAI*, 2020, pp. 668–678.

Books and Chapters

1. S. Liu, D. Guo, Z. Liu, **T. Wang**, Q. Qin, X. V. Wang, and L. Wang, "A digital twin-enabled approach to reliable human–robot collaborative assembly", in *Human-Centric Smart Manufacturing Towards Industry 5.0*, B. Wang, P. Zheng, L. Wang, and D. Mourtzis, Eds., Springer, 2025, pp. 281–304. DOI: 10.1007/978-3-031-82170-7_12.

Preprints and Working Papers

1. Z. Liu, **T. Wang**, X. V. Wang, and L. Wang, *Agentic ERP: Multi-agent large language model architecture for autonomous enterprise resource planning*, arXiv:2607.17331, Preprint, 2026. URL: https://arxiv.org/abs/2607.17331.
2. **T. Wang**, M. Li, X. V. Wang, and L. Wang, *SNR-adaptive local posterior correction for frozen tabular predictors*, SSRN 7478518, Preprint, 2026. URL: https://papers.ssrn.com/sol3/papers.cfm?abstract_id=7478518.
3. **T. Wang**, X. V. Wang, L. Wang, M. Li, and Z. Liu, *The anatomy and boundary of adaptation under temporal tabular shift*, arXiv:2609.12136, Preprint, 2026. URL: https://arxiv.org/abs/2609.12136.
4. **T. Wang**, Y. Wang, Z. Liu, X. V. Wang, and L. Wang, *Ghost attractor networks: Basin-structured dynamical decoders for closed-loop sequential generation*, arXiv:2606.18315, Preprint, 2026. URL: https://arxiv.org/abs/2606.18315.
5. **T. Wang**, M. Li, X. V. Wang, and L. Wang, *Amortized posterior inference over correction behaviors for post-hoc tabular prediction*, SSRN 6946590, Preprint; publication date not specified. URL: https://papers.ssrn.com/sol3/papers.cfm?abstract_id=6946590.

Skills

Languages	Strong reading, writing and speaking competencies for English and Mandarin Chinese.
Coding	Python, Matlab, R, C/C++, Java, JavaScript, Swift, HTML/CSS, React, SQL, LaTeX, ...
Frameworks	PyTorch, Ollama, RAGFlow, Tensorflow, Keras, Next.js, Unity, RobotStudio, ...
Misc.	Independent research, teaching, training, presenting, and publishing.

Miscellaneous Experiences

Awards and Achievements

2024	Graduate National Scholarship
	Outstanding Graduate of Shanghai Jiao Tong University (SJTU)
	KLA Scholarship of UM-SJTU Joint Institute
	SJTU-KTH Seed Grant
	Three Good Student of SJTU
2023	Third Prize at Ph.D. Academic Forum of SJTU
2021	Ya Dong scholarship of UM-SJTU Joint Institute
2020	Zhi Yuan Honorary Ph.D. student of SJTU
2019	Sam and Daisy Wu scholarship of UM-SJTU Joint Institute
2018	Second Prize for Band C in National English Competition for College Students
2017	"Honorable Mention" in International Interdisciplinary Contest in Modeling (ICM)

Certification

2024	Elsevier Reviewer for Journal of Manufacturing Systems
	Graduate Recruitment Ambassador for UM-SJTU Joint Institute
2023	Reviewer for Medical Image Computing and Computer Assisted Intervention (MICCAI) Society
2022	President of SJTU Photography Association
2019	Volunteer for the Boston Food Pantry
2018	Outstanding Performance in Volunteering Activities and Education in Yunnan Province
	Volunteer for Shanghai International Marathon
	CET-4 and CET-6 English Language Certificates